BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Flowcharts and Algorithm Design

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. For cost = price x quantity, identify input, processing and output.

2. What decides whether an action is legal?

A) Its popularity
B) Its speed
C) Its cost
D) The law

3. What is a privacy concern with collecting personal data?

4. An online service uses only one language. "Technology cannot affect language or who takes part." Correct this. Who might be left out?

Today's objectives

Recognise the purpose of standard OCR J277 flowchart symbols. Trace a Yes/No decision for two values.

Key vocabulary - Read, cover, recall

Flowchart
Diagram of algorithm steps and control flow.

Terminal
Start or End symbol.

Decision
Diamond evaluating a condition.

Process
Rectangle for a calculation or instruction.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Reading a flowchart	A flowchart represents an algorithm visually. Follow arrows from Start through each process or decision until End.
Use the right shapes	Start and End use terminal shapes. Inputs and outputs use parallelograms, processes use rectangles, and decisions use diamonds. Predefined subprograms have their own symbol.
Decisions choose one branch	A diamond evaluates a condition. True/Yes and False/No lead down alternative routes, so a value follows only the appropriate path.
Correct loops and missing paths	Follow the algorithm with a test value. If input is invalid, an arrow can return to input/check for another attempt. Each branch should lead to a defined next action or End.
Structure diagrams	A structure diagram shows decomposition: Quiz has Menu, Play and Results beneath it; Play has Input and Scoring beneath it. Links show the hierarchy between the subproblems. Unlike a flowchart, this diagram does

I Do - One worked example

A flowchart checks if a mark is at least 5. If true it prints 'Pass'; otherwise it prints 'Try again'. Explain which branch is followed for 7 and 3. [4 marks]

1. The diamond checks mark >= 5. Only one branch is followed per test.

2. 7 >= 5 is true [1], so the Yes branch prints Pass [1].

3. 3 >= 5 is false [1], so the No branch prints Try again [1].

4. Check both comparisons and the different outputs; avoid running both branches for one input.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A flowchart checks whether a number is greater than 10. Explain the route and output for 12 and 8. [4 marks]

2. A lunch-order flowchart repeats a question after an invalid choice. Explain why an arrow loops back to the input. [4 marks]

3. A flowchart uses a rectangle 'total = a + b' then a parallelogram 'Display total'. Explain both symbols. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict whether this image file fits.

```
freeMB = 100
imageMB = 80
if imageMB <= freeMB then
print("Fits")
else
print("Too large")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State which flowchart shape represents a decision. [1 mark]

Q3. A flowchart checks if a customer is at least 16 years old before printing Eligible. Explain what happens for 18 and 14. [4 marks]

Q4. Describe the purposes of terminal, input/output, process and decision symbols in flowcharts and the use of arrows. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A flowchart's decision diamond can direct execution down one of two branches. ☐ T ☐ F
2. A flowchart represents an algorithm visually ☐ T ☐ F
3. Start and End use terminal shapes ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Inputs: price and quantity. Processing: multiplication. Output: total cost. ☐

Q2 D - The law ☐

Q3 People may lose control over how identifying information is used or shared. ☐

Q4 Technology can change language, access, traditions and who can take part. Consider who benefits or is harmed. Something legal may still be unfair. ☐

Code First - Model

```
freeMB = 100
imageMB = 80
if imageMB <= freeMB then
print("Fits")
else
print("Too large")
endif
```

Output: Fits

We Do - Check the model

1. 12 > 10 is true [1], follows Yes and prints High [1]. 8 > 10 is false [1], follows No and prints Low [1]. ☐

2. Invalid choice follows the No route [1] and causes another input request [1]. The return arrow rechecks the new choice [1] until it is valid [1]. ☐

3. Rectangle represents processing [1] calculating and storing a sum [1]. Parallelogram represents output [1], displaying the calculated total [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 A diamond [1]. ☐

Q3 18 >= 16 is true [1] so Eligible prints [1]. 14 >= 16 is false [1], so the not-eligible alternative prints [1]. ☐

Q4 Terminal identifies start/end [1]; parallelogram input/output [1]; rectangle performs process/calculation [1]; diamond checks condition [1]; it directs to alternative Yes/No branches [1]; arrows show sequence/control flow [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: